Zaawansowane techniki programowania

Laboratorium 7

ZMIENDAK WIKTOR

ZADANIA OBOWIĄZKOWE:

1. Uzupełniamy RestProductClientAdapter tworząc walidację dla getProduct oraz reserve.

```java
public RestProductClientAdapter(RestProductGateway gateway) {
    this.gateway = Objects.requireNonNull(gateway);
}

@Override
public ProductView getProduct(long productId) {
    try {
        return gateway.fetchProduct(productId);
    } catch (RestNotFoundException e) {
        throw new NoSuchElementException(s: "Product not found");
    }
}

@Override
public void reserve(long productId, int quantity) {
    try {
        gateway.reserve(productId, quantity);
    } catch (RestNotFoundException e) {
        throw new NoSuchElementException(s: "Product not found");
    } catch (RestConflictException e) {
        throw new IllegalStateException(s: "Insufficient stock");
    }
}
```

2. Uzupełniamy GrpcProductClientAdapter i tworzymy walidację dla getProduct oraz reserve podobnie jak w poprzednim podpunkcie.

```java
public GrpcProductClientAdapter(GrpcProductGateway gateway) {
    this.gateway = Objects.requireNonNull(gateway);
}

@Override
public ProductView getProduct(long productId) {
    try {
        return gateway.getProduct(productId);
    } catch (GrpcStatusException e) {
        if (e.getCode() == GrpcStatusCode.NOT_FOUND) {
            throw new NoSuchElementException(s: "Product not found");
        }
        throw e;
    }
}

@Override
public void reserve(long productId, int quantity) {
    try {
        gateway.reserve(productId, quantity);
    } catch (GrpcStatusException e) {
        if (e.getCode() == GrpcStatusCode.NOT_FOUND) {
            throw new NoSuchElementException(s: "Product not found");
        }
        if (e.getCode() == GrpcStatusCode.FAILED_PRECONDITION) {
            throw new IllegalStateException(s: "Insufficient stock");
        }
        throw e;
    }
```

3. Uzupełniamy ProductClientFactory tak była możliwość wyboru klienta (REST lub GRPC).

```
public ProductClient create(AppConfig cfg) {
    if (cfg.mode() == AppConfig.Mode.REST) {
        return new RestProductClientAdapter(restGateway);
    }
    if (cfg.mode() == AppConfig.Mode.GRPC) {
        return new GrpcProductClientAdapter(grpcGateway);
    }
    throw new IllegalArgumentException("Unsupported mode: " + cfg.mode());
}
```

4. Uzupełniamy OrderService tworząc placeOrder z walidacją ilości zamówień.

```
public Order placeOrder(long productId, int quantity) {
    if (quantity <= 0) {
        throw new IllegalArgumentException(s: "Quantity must be > 0");
    }
    ProductView product = productClient.getProduct(productId);
    productClient.reserve(productId, quantity);
    return new Order(idSeq.incrementAndGet(), productId, quantity, total(product, quantity), now());
}
```

ZADANIA DODATKOWE:

1. Uzupełniamy RetryingProductClient

```
@Override
public ProductView getProduct(long productId) {
    for (int attempt = 1; attempt <= maxAttempts; attempt++) {
        try {
            return delegate.getProduct(productId);
        } catch (ServiceUnavailableException e) {
            if (attempt == maxAttempts) {
                throw new IllegalStateException(s: "Product service unavailable");
            }
        } catch (NoSuchElementException | IllegalStateException e) {
            throw e;
        }
    }
    throw new IllegalStateException(s: "Product service unavailable");
}

@Override
public void reserve(long productId, int quantity) {
    for (int attempt = 1; attempt <= maxAttempts; attempt++) {
        try {
            delegate.reserve(productId, quantity);
            return;
        } catch (ServiceUnavailableException e) {
            if (attempt == maxAttempts) {
                throw new IllegalStateException(s: "Product service unavailable");
            }
        } catch (NoSuchElementException | IllegalStateException e) {
            throw e;
        }
    }
    throw new IllegalStateException(s: "Product service unavailable");
}
```

2. Uzupełniamy CachingProductClient

```java
@Override
public ProductView getProduct(long productId) {
    return cache.computeIfAbsent(productId, delegate::getProduct);
}

@Override
public void reserve(long productId, int quantity) {
    delegate.reserve(productId, quantity);
    cache.remove(productId);
}
```

3. W ProductClientFactory dodajemy pipeliny dekoratorów.

```java
public ProductClient create(AppConfig cfg) {
    ProductClient client;
    if (cfg.mode() == AppConfig.Mode.REST) {
        client = new RestProductClientAdapter(restGateway);
    } else if (cfg.mode() == AppConfig.Mode.GRPC) {
        client = new GrpcProductClientAdapter(grpcGateway);
    } else {
        throw new IllegalArgumentException("Unsupported mode: " + cfg.mode());
    }

    if (cfg.retryEnabled()) {
        client = new RetryingProductClient(client, cfg.maxAttempts());
    }
    if (cfg.cacheEnabled()) {
        client = new CachingProductClient(client);
    }
    return client;
}
```

WYNIKI TESTÓW

```
[INFO] --- compiler:3.15.0:compile (default-compile) @ lab7-patterns-productclient ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ lab7-patterns-productclient ---
[INFO] skip non existing resourceDirectory C:\Users\ziggs\Desktop\Studies\Cybersecurity\Semestr_I\ZTP\Zmiendak_Wiktor_LAB7\Zmiendak_Wiktor_LAB7\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ lab7-patterns-productclient ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ lab7-patterns-productclient ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running pl.example.lab7.Grade3AdaptersMappingTest
WARNING: A Java agent has been loaded dynamically (C:\Users\ziggs\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.12\byte-buddy-agent-1.14.12.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.987 s -- in pl.example.lab7.Grade3AdaptersMappingTest
[INFO] Running pl.example.lab7.Grade3FactorySelectionTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.037 s -- in pl.example.lab7.Grade3FactorySelectionTest
[INFO] Running pl.example.lab7.Grade3OrderServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.207 s -- in pl.example.lab7.Grade3OrderServiceTest
[INFO] Running pl.example.lab7.Grade4CachingProductClientTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.036 s -- in pl.example.lab7.Grade4CachingProductClientTest
[INFO] Running pl.example.lab7.Grade4FactoryPipelineTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.033 s -- in pl.example.lab7.Grade4FactoryPipelineTest
[INFO] Running pl.example.lab7.Grade4RetryingProductClientTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.058 s -- in pl.example.lab7.Grade4RetryingProductClientTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 15, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  13.420 s
[INFO] Finished at: 2026-04-26T10:46:01+02:00
[INFO] ------------------------------------------------------------------------
```